PUBLICATION

Kai-zen (改善) The Case for Continuous Model Improvement

Why static AI models fail in medicine, and how ClinixSummary's weekly update cycle and clinician feedback loop produce consistently improving outputs.

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1. Executive Summary

Most AI systems are trained once and deployed as static models, receiving updates infrequently — sometimes quarterly, sometimes annually. In clinical documentation, this approach is fundamentally flawed. Medical language evolves, clinical guidelines change, new medications enter the market, and individual clinician preferences vary.

2. Why Static Models Fail in Medicine

- New medications, procedures, and clinical guidelines emerge continuously. A model trained in January lacks awareness of a drug approved in March.
- Medical terminology evolves — diagnostic criteria change (e.g., sepsis definitions), classification systems are updated (ICD-10 revisions), and new clinical concepts emerge.
- Clinician feedback reveals systematic patterns — repeated corrections indicate model deficiencies that should be addressed, not endured.
- Specialty expansion requires continuous learning — as ClinixSummary adds new specialties, models must incorporate new domain knowledge.

3. The Kai-zen Process

3.1 Feedback Collection

Every clinician interaction with a ClinixSummary note generates potential learning signal: edits, corrections, section rearrangements, terminology substitutions, and explicit ratings. This feedback is captured, de-identified, and aggregated automatically.

3.2 Pattern Analysis

Aggregated feedback is analysed weekly to identify systematic patterns: consistently mis-recognised terms, section classification errors specific to certain specialties, new terminology appearing in clinician corrections, and accuracy variations across languages or regions.

3.3 Prioritised Retraining

Identified patterns are prioritised by clinical impact and frequency. High-impact issues (medication names, dosage recognition) are expedited. Retraining targets the specific model components affected, avoiding unnecessary changes to well-performing areas.

3.4 Validation and Deployment

Every model update passes through the Clinix QM validation gates: automated regression testing, specialty benchmarks, clinical review, and A/B testing before full deployment. Updates that do not meet or exceed the previous version’s performance are rejected.

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4. Results: Measurable Improvement Over Time

ClinixSummary's Clinician Acceptance Rate has improved consistently since launch:

- Q1 2025 (launch): 87% average acceptance rate.
- Q2 2025: 90% (+3 points).
- Q3 2025: 92% (+2 points).
- Q4 2025: 93.5% (+1.5 points).
- Target Q2 2026: >95%.

Average edits per note have decreased from 4.2 at launch to 1.8 — a 57% reduction in clinician correction effort.

5. Conclusion

Continuous improvement is not a feature — it is a design philosophy that recognises the dynamic nature of medicine. ClinixSummary's Kai-zen approach ensures that the system gets better every week, driven by the collective intelligence of the clinicians who use it.

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